

## **ANNEX**

### **DRAFT GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE**

#### **I. INTRODUCTION**

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as “the 2021 Recommendation”) that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i. e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO’s Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

#### **II. WHAT ARE THE AIMS OF THIS CONSULTATION?**

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

#### **III. HOW TO FILL IN THE QUESTIONNAIRE?**

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on

the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at <https://www.unesco.org/en/open-science/toolkit> and <https://unesdoc.unesco.org/ark:/48223/pf0000387324>.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

- (i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted online.
- (ii) Email: the questionnaire can be completed electronically and sent to: [openscience@unesco.org](mailto:openscience@unesco.org)

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

#### IV. DRAFT QUESTIONNAIRE

##### A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country\*: **Switzerland**
- Organization(s) or entity(s) responsible for the preparation of the report\*:  
Name: **EPFL (Swiss Federal Institute of Technology Lausanne)—Open Science Domain**  
Website: <https://www.epfl.ch/research/open-science/>  
Email address\*:   
Phone number\*:   
Please describe the role/mandate of your organization:

**EPFL is one of two federally funded higher education institutions of Switzerland and is a major research actor. Its Open Science Domain supports Open Science both within the institution and in national bodies (Swissuniversities Delegation Open Science, ORD Strategy Council, ETH Domain ORD Programme)**

- Officially designated contact person(s) who completed this survey:

Full name\*: **Dr Gilles Dubochet**

Position\*: **Head of Open Science** Email

address\*:

- Full Name(s) of designated official(s) certifying the report:

**Christof Bareiss, Federal Department of Foreign Affairs (FDFA)**

- Brief description of the consultation process established for the preparation of the report:  
**This report was informed by ongoing and regular exchanges with key stakeholders. No specific consultations were conducted solely for the purpose of responding to this survey.**
- Other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey: **None**

Name of the organization\*:

Website\*:

Email address\*:

Sector: ☐ Public ☐ Private ☐ Civil Society ☐ Other

## **B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION**

### **1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE**

- 1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

**YES** / NO

**If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

**The Recommendation was shared and discussed in swissuniversities—the umbrella organization of the Swiss higher education institutions—through which the Recommendation was widely accessible.**

- 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES / NO / **PARTLY**

**If yes**, which language(s)? **The Recommendation is available in French, one of Switzerland's national languages (spoken by approximately 22% of the population) but not in German (62%) or Italian (8%).**

- 1.3 Have there been awareness raising activities on open science, including all its key elements<sup>i</sup>, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities?

(ref.: [\(i\) 16.f, 16.g, 16.h](#))

**YES** / NO

**If yes**, please provide details on the activities organized or foreseen and links as relevant.

- **In January 2025, swissuniversities, hosted an event to review the progress, achievements, and challenges in Switzerland of the Open Science Programme,**

which is dedicated to implementing the National Strategies for OA and ORD. The event included reflections on the first phase of the programme (2021-2024) and an outlook on future initiatives:

<https://www.swissuniversities.ch/en/news-1/events/invitation-programme-open-science-event-22-january-2025>

- March 2025: within the mandate of the ETH Domain Open Research Data Programme, lead representative of the Center for Information Technology, Society, and Law (hosted by the University of Zurich) will present the main findings of a research study that analyzed the potential legal hurdles at the federal level in Switzerland for publishing research data in openly accessible formats. An ORD guideline template that was drafted based on this analysis will be shared during the event, to be implemented by the six ETH Domain Institutions: PSI (Paul Scherrer Institute), WSL (Swiss Federal Institute for Forest, Snow and Landscape Research), Empa (Swiss Federal Laboratories for Materials Science and Technology), Eawag (Swiss Federal Institute of Aquatic Science and Technology), ETHZ (Swiss Federal Institute of Technology Zurich), and EPFL (Swiss Federal Institute of Technology Lausanne):

<https://memento.epfl.ch/event/open-research-data-legal-limits-and-design-options/>

- In April 2025, the Annual Symposium of the Swiss ERIC (European research infrastructure consortia) community, organized by the State Secretariat for Education, Research and Innovation, will explore the overarching theme “Research Infrastructures as Enablers of Open Science”. It will offer a platform for dialogue between the research communities and research & innovation partners that play a role in Switzerland’s participation in European research infrastructure networks:

<https://e2-news.ch/en/news/annual-symposium-of-swiss-eric-community>

- June 2025: The ETH Domain Measure 1 Workshop, “Advancing Open Research Data Practices”, will showcase the outcomes of Measure 1—one of five Measures (see 2.3) designed to enhance ORD practices and promote open science across ETH Domain researchers and institutions (EPFL, ETHZ, PSI, WSL, Empa, and Eawag). The primary goal of Measure 1 was to support them in developing and leading ORD practices in their fields. The event will feature presentations of funded ORD projects and interactive workshops on key ORD topics:

<https://ethrat.ch/en/the-five-measures-in-details/>

Notable activities from 2024 include:

- The Swiss National Science Foundation’s event, “Research on Research,” showcased key relevant national and international projects and facilitated discussions on advancing the field. Research on Research initiatives play a crucial role in shaping an evidence-based scientific system, particularly in research funding and evaluation. This event aimed to highlight relevant projects while fostering community building in Switzerland.
- Open Science Days, sponsored by swissuniversities and the Swiss National Science Foundation (SNSF), Switzerland’s primary scientific research funder. Four Swiss academic institutions—UNIL (University of Lausanne), HES-SO (University of Applied Sciences and Arts Western Switzerland), HEP Vaud (University of Teacher Education Vaud), and EPFL (Swiss Federal Institute of Technology Lausanne)—collaborated to organize a week-long event featuring

key elements of Open Science (OS): Open Access (OA), Open Research Data (ORD), Open Educational Resources (OER), and citizen science. The event also explored the benefits and challenges of these elements, as well as the impact of artificial intelligence on research and education:

<https://opensciencedays.ch/en/>

- 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science<sup>ii</sup>, in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

**YES** / NO

**If yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?

- The Swiss National Science Foundation (SNSF), through the Swiss National Open Access Strategy (see 2.2), mandates full, immediate, and free access to publications resulting from publicly funded research:

<https://www.snf.ch/en/VyUvGzptStOEPUoC/topic/open-access-publications>

- The Swiss National Open Research Data (ORD) Strategy (see 2.2) mandates the application of the FAIR (Findable, Accessible, Interoperable and Reusable) principles to publicly financed research data:

<https://www.swissuniversities.ch/en/topics/open-science/open-research-data/national-strategy>

## 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

- 2.1 Does your country have a national policy<sup>iii</sup>, strategy or plan of action on science, technology and innovation (STI)?

YES / NO / **PARTIALLY**

**If yes**

- please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*
- does it include any of the following open science elements:
  - open access to scientific knowledge: **YES**
  - open science infrastructure: **YES**
  - open engagement of societal actors: **NO**
  - open dialogue with other knowledge systems: **NO**

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There is no single policy on STI in Switzerland, but several policy instruments at the Federal and Cantonal level, as well as by the delegated entities for education, research, and innovation policy in Switzerland: swissuniversities, the ETH Board, the Swiss National Science Foundation (SNSF), and the Swiss Academies of Science.

At the Federal level, the main instrument is the “Federal Act on the Promotion of Research and Innovation” (RIPA) (enclosed pdf) , operationalized with multi-year strategic plans—Education, Research, and Innovation (ERI) Dispatches—that periodically define strategic priorities and funding allocations for STI:

<https://www.fedlex.admin.ch/eli/cc/2013/786/en>

<https://www.sbf.admin.ch/sbf/en/home/eri-policy/eri-21-24.html>

- Authority in charge of development: The State Secretariat for Education, Research and Innovation (SERI), which oversees national research and innovation policy.
- Entities involved and the process of development: The SERI is responsible for the development and implementation of the Act. It has designated the SNSF, the Swiss Academies of Arts and Sciences, and Innosuisse—the Swiss Innovation Agency—as key institutions for promoting the Act, supported through federal subsidies.
- Year of Adoption: Decision in December 2012, in force January 2014
- Year of last update: July 2023
- Key areas: principles and missions, open science and transparency, legal and administrative provisions, research promotion by the federal administration, innovation support, compliance with scientific integrity and good scientific practice, exploitation of research findings, international cooperation in research and innovation, financing, research and innovation policy planning, information and reporting obligations, quality control.

2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)<sup>iv</sup> at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

**YES** / PARTLY / NO

**If yes or partly**, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

The two main policies are:

(i) The Swiss National Open Access Strategy (enclosed pdf):

[https://www.swissuniversities.ch/fileadmin/swissuniversities/Dokumente/Hochschulpolitik/Open\\_Access/Swiss-National-Open-Access-Strategy-2024-en.pdf](https://www.swissuniversities.ch/fileadmin/swissuniversities/Dokumente/Hochschulpolitik/Open_Access/Swiss-National-Open-Access-Strategy-2024-en.pdf)

- Authority in charge of development: swissuniversities and the Swiss National Science Foundation (SNSF), as entrusted by the State Secretariat for Education, Research, and Innovation (SERI).
- Entities involved and the process of development: swissuniversities and the SNSF.
- Year of adoption: the original OA strategy was adopted in 2017
- Year of last update: a revised OA strategy was adopted by swissuniversities and SNSF in May 2024.
- Key areas: the revised OA strategy focuses on its mandate, scope, vision, and international policy context. It is guided by principles such as diversity in scholarly publishing, academic quality, financial sustainability, international

collaboration, institutional support, and broad disciplinary engagement. Core objectives include developing and connecting research infrastructures, negotiating with service providers for a sustainable transformation, establishing supportive legal frameworks, promoting OA for diverse publication formats, expanding research assessment criteria, and monitoring OA adoption and financial impacts. Effective governance will ensure the successful implementation of the strategy.

(ii) The Swiss National ORD Strategy (enclosed pdf):

<https://www.swissuniversities.ch/en/topics/open-science/open-research-data/national-strategy#:~:text=ORD%20Action%20Plan,-Based%20on%20the&text=The%20Action%20Plan%20covers%20the,for%20the%20ORD%20Action%20Plan>  
[https://www.swissuniversities.ch/fileadmin/swissuniversities/Dokumente/Hochschulpolitik/ORD/Swiss\\_National\\_ORD\\_Strategy\\_en.pdf](https://www.swissuniversities.ch/fileadmin/swissuniversities/Dokumente/Hochschulpolitik/ORD/Swiss_National_ORD_Strategy_en.pdf)

- Authority in charge of development: swissuniversities (entrusted by the SERI)
- Entities party to the Strategy: swissuniversities, the ETH Board, the SNSF, and the Swiss Academies of Arts and Sciences.
- Year of adoption: 2021
- Year of last update: the ORD Strategy has been consolidated further by an ORD Action plan, adopted in 2022 (see 2.4).
- Key areas: the ORD Strategy outlines its mandate, scope, vision, and broader context, guided by principles such as FAIR principles, disciplinary diversity, national and international integration, and sustainability. Its key objectives include supporting researchers, research communities, and institutions in adopting ORD practices; developing, promoting, and maintaining ORD infrastructures and services; fostering skills development and best practice exchange; and ensuring effective governance for its implementation.

2.3 In your country, are there specific policy instruments<sup>v</sup> that aim to promote open science?

**YES/ UNDER DEVELOPMENT/NO**

**If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

The two main policy instruments are:

(i) Federal project contributions (PgB) Open Science Programmae II (2025-2026):

<https://www.swissuniversities.ch/en/topics/open-science/open-science-programme>

- Objective: To strategically develop Open Science in Swiss universities and research communities based on the Swiss National Strategies for OA and ORD, as well as other innovative areas of open science, in particular OS-compatible knowledge and tech-transfer between HEIs and industrial and societal stakeholders.
- Authority in charge: swissuniversities
- Allocated budget: CHF 6.45 million
- Elements of open science: open access, open research data, innovative areas for open science

(ii) The ETH Domain Open Research Data Program

<https://ethrat.ch/en/eth-domain/open-research-data/>

- Objective: With the broader aim of advancing Open Science among ETH Domain institutions the Program defines five measures:



- Measure 1: support researchers in developing and leading ORD practices
- Measure 2: improve the interoperability and access to the research data management (RDM) ecosystem
- Measure 3: RDM training
- Measure 4: Legal aspects of RDM
- Measure 5: Careers in ORD practices.
- Authority in charge: the ETH Board with the 5 ETH Domain institutions
- Allocated budget: CHF 15 million
- Elements of open science: please refer the five measures (above)

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (*ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h*)

**YES** / UNDER DEVELOPMENT / NO

**If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

The Action Plan for ORD, developed under the leadership of swissuniversities and in collaboration with SNSF, ETH Domain and science academies, concretizes the measures of the National ORD Strategy adopted in 2021. Based on the objectives set forth in the ORD Strategy, four action areas are defined in the Action Plan:

- i) Support for researchers, research communities, and institutions on adopting ORD practices.
- ii) Development, promotion, and maintenance of ORD infrastructures and services.
- iii) ORD skills development and exchange of best practices.
- iv) Building up systemic and supportive conditions: incentives and rewards, ethical and legal aspects, international alignment.

The Swiss National ORD Strategy Council—a new governance entity uniting the senior office holders of these institutions—coordinates the implementation of the Action Plan, with a specific focus on open science research infrastructures. In addition, the Swiss ORD community—two sounding boards, consisting of researchers and service providers from the ERI institutions—also supports the strategic development and implementation of ORD measures.

Links:

[https://www.swissuniversities.ch/fileadmin/swissuniversities/Dokumente/Hochschulpolitik/ORD/ActionPlanV1.0\\_December\\_2021\\_def.pdf](https://www.swissuniversities.ch/fileadmin/swissuniversities/Dokumente/Hochschulpolitik/ORD/ActionPlanV1.0_December_2021_def.pdf)  
<https://openresearchdata.swiss>

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (*ref.: (ii) 17.d, 17.e, 17.g*)

**YES / UNDER DEVELOPMENT** / NO

**If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and [elements of open science](#) **Error! Bookmark not defined.** included in the policy (please consider p



olicies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

Swiss institutions have implemented comprehensive policies embodying key elements of open science that promote OA to research outputs, foster collaboration, and enhance transparency. For example:

Universities:

- With the implementation of its open science policy in 2021, the University of Zurich fosters an open academic culture that embodies the National OA Strategy of swissuniversities (see 2.2) and the Open Science Roadmap of the League of European Research Universities (LERU). The policy prioritizes three areas of action: (i) public and free access to scientific resources (e.g., data, codes, and publications); (ii) improved research methods that reflect the reproducibility, replicability, transparency and efficiency of scientific results; and (iii) a cultural change in university structures that includes changes to assessment of scientific achievements:

<https://www.uzh.ch/en/researchinnovation/ethics/openscience.html>

- The University of Bern actively promotes open science through a range of institutional services and infrastructures. These include support for OA publishing, research data management, and long-term data preservation. Additionally, a dedicated open science team, in collaboration with the university library, organizes regular courses and workshops on open science-related topics for the university's researcher community:

[https://www.ub.unibe.ch/services/open\\_science/index\\_eng.html](https://www.ub.unibe.ch/services/open_science/index_eng.html)

Universities of Applied Sciences/ "Haute École Spécialisée" (HES):

- Since 2018, the Zurich University of Applied Sciences (ZHAW) has promoted open science. It supports the National OA Strategy through an institutional OA policy and managing an institutional repository through its University Library. It also advances the Swiss National ORD Strategy with research data services and training. Additionally, ZHAW fosters Open Educational Resources (OERs) as part of the Education and Digital Transformation strategy (2018–2029) and the ZHAW OER Policy, recommending that teaching materials be published under free licenses like CC BY or CC BY-SA:

<https://www.zhaw.ch/de/fokusthemen/open-science>

- University of Applied Sciences and Arts of Western Switzerland (HES-SO): launched in December 2018, the Open HES-SO Strategy aims to make all HES-SO research publications and data openly accessible. The Open Science HES-SO Community, established in October 2020, now includes around 90 members across various fields and roles (researchers, professors, librarians, publishers, lawyers, scientific assistants, etc.). It serves as a hub for reflection, experience-sharing, and action on Open Science, organizing quarterly plenary sessions and dedicated working groups on key topics:

<https://www.hes-so.ch/recherche-innovation/open-science>

[https://www.hes-so.ch/fileadmin/documents/HES-SO/Documents\\_HES-SO/pdf/open-science/2018-Strategie-Open-HES-SO-12321.pdf](https://www.hes-so.ch/fileadmin/documents/HES-SO/Documents_HES-SO/pdf/open-science/2018-Strategie-Open-HES-SO-12321.pdf)

ETH Domain:

- In 2019, EPFL (Swiss Federal Institute of Technology Lausanne) adopted its OA policy, requiring all research results to be published in OA, with mandatory deposit in the institutional repository. This policy aligns with EPFL's broader Open Science strategy, which fosters a cultural shift driven by researchers and research communities. The evolving strategy outlines EPFL's path toward institutional change by promoting open communities, open infrastructure, open knowledge, and societal engagement. By ensuring maximum visibility for research results, the OA policy strengthens this framework, advancing knowledge dissemination, outreach, and the development of open infrastructure and communities:

[https://www.epfl.ch/about/overview/wp-content/uploads/2019/09/LEX-3.5.1\\_EN.pdf](https://www.epfl.ch/about/overview/wp-content/uploads/2019/09/LEX-3.5.1_EN.pdf)

<https://www.epfl.ch/research/open-science/>

- Since 2018, ETHZ (Swiss Federal Institute of Technology Zurich) implemented its OA policy, requiring researchers to make all research outputs publicly accessible via the institutional repository:

<https://rechtssammlung.sp.ethz.ch/Dokumente/134.pdf>

<https://ethz.ch/en/research/open-science.html>

2.6 In your country, are there specific funding mechanisms<sup>vi</sup> for open science at the national level? (ref.: (ii) 17.j. (iii) 18.j. (iv) 19.c)

**YES** / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

Switzerland has established specific national funding mechanisms to support open science initiatives:

- The funding scheme Open Science Programme II (2025-2026)—a continuation of its predecessor, Open Science Programme I (2021-2024)—is based on the Swiss National Strategies for OA and ORD. It supports developments along the axes of OA, ORD, and innovative areas for open science through knowledge transfer between various stakeholders in academia, industry, and society:

<https://www.swissuniversities.ch/en/topics/open-science/open-science-programme>

- The ETH Domain ORD Program financially supports researchers across six Swiss public research institutions—PSI (Paul Scherrer Institute), WSL (Swiss Federal Institute for Forest, Snow and Landscape Research), Empa (Swiss Federal Laboratories for Materials Science and Technology), Eawag (Swiss Federal Institute of Aquatic Science and Technology), ETHZ (Swiss Federal Institute of Technology Zurich), and EPFL (Swiss Federal Institute of Technology Lausanne)—to advance ORD practices. The program funds five key measures (see also 2.3): (i) field-specific actions, (ii) coordinating RDM services, (iii) developing RDM training courses, (iv) addressing legal RDM issues, and (v) creating career paths for ORD professionals. Funds are awarded along “Establish” (links existing ORD practices to broader research agendas), “Explore” (tests early-stage practices), and “Contribute” (integration into existing, often global infrastructures) axes:

<https://open-research-data-portal.ch/projects/>

<https://ethrat.ch/en/the-five-measures-in-details/>

- In line with the Swiss National Science Foundation (SNSF)'s mandate for full and immediate access to publicly funded research outputs (see also 2.2), it provides financial support for OA publishing of research articles, books, and book chapters. These funds cover publisher's services for quality assurance, production and distribution, article processing charges, and proofreading. Additionally, the SNSF funds other aspects of open science, including open dialogue between scientists and society as well as science communication training for researchers:  
<https://www.snf.ch/en/A7fep1IPxz1XezVS/page/find-funding-scheme?category=DQkA46U4rzqBqIBR%3B&page=0>

2.7 In your country, have open science practices<sup>vii</sup> been integrated in existing research funding mechanisms?

**YES** / NO

If **yes**, please provide details and links as relevant.

Most funding instruments in Switzerland include requirements concerning Open Access and Open Research Data. For example, the SNSF requires all publicly funded research outputs to be publicly and freely accessible:  
<https://www.snf.ch/en/dah3uC2QX95tfPNd/topic/open-science>

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (*ref.: (ii) 17.i*)

YES / **NO**

If **yes**, please provide details and links as relevant.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

While specific exchanges with private actors on open science practices exist, they have not been structured into a coherent national framework concerning equitable public-private partnerships on open science.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (*ref.: (ii) 17.j, 23.c*)

YES / **UNDER DEVELOPMENT** / NO

If **yes** or **under development**, please provide details and links as relevant.

The Swiss Open Access Monitor is the national framework for monitoring of Open Access to scientific publications. The monitor provides basic indicators for visualizing the progress of the Swiss National OA Strategy across two types of resources: OA journals and OA repositories. It was developed within the framework of the National Open Access Monitoring (NOAM) project, led by the Consortium of Swiss Academic Libraries, and funded by the Open Science programme 2021-2024 and the Consortium. This framework is further being developed.

Link: <https://oamonitor.ch>

There is no national monitoring framework when it comes to Open Research Data, although SNSF has been monitoring uptake of ORD across its funding schemes:

Link: <https://data.snf.ch/stories/open-research-data-2023-en.html>

### 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

- 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

The latest official data indicates that Switzerland dedicated CHF 24.6 billion to R&D in 2021, representing 3.3% of its GDP (6<sup>th</sup> in international rankings). This represents an average annual increase of 4% compared with 2019—the year of the last survey. This data was published in March-May 2023 by the Federal Statistical Office. Government budget allocations for R&D in 2023 are not yet published:

<https://www.bfs.admin.ch/bfs/en/home/statistics/education-science/technology.html>

[https://www.admin.ch/gov/en/start/documentation/media-releases.msg-id-95387.html?utm\\_source=chatgpt.com](https://www.admin.ch/gov/en/start/documentation/media-releases.msg-id-95387.html?utm_source=chatgpt.com)

- 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

Switzerland has one of the world's most advanced internet infrastructures. As of early 2024, 99% of the population had internet access, up from 97% in 2023, according to DATAREPORTAL (February 2024):

<https://datareportal.com/reports/digital-2024-switzerland#:~:text=Switzerland's%20internet%20penetration%20rate%20stood,January%202023%20and%20January%202024.>

As of November 2023, the Swiss Federal Communications Commission reported Switzerland's strong broadband infrastructure. By the end of 2022, 48.2% of the population had high-speed internet access, surpassing France (46.6%), Norway (45.7%), Korea (45.4%), and the OECD average (34.9%).

As of March 2024, Switzerland ranked 14th globally in broadband speeds, with a median transfer rate of 205.06 Mbit/sec:

[https://en.wikipedia.org/wiki/Internet\\_in\\_Switzerland#cite\\_note-1-2](https://en.wikipedia.org/wiki/Internet_in_Switzerland#cite_note-1-2)

[https://en.wikipedia.org/wiki/List\\_of\\_countries\\_by\\_Internet\\_connection\\_speeds#Mobile\\_connection](https://en.wikipedia.org/wiki/List_of_countries_by_Internet_connection_speeds#Mobile_connection)

- 3.3 Are there national research and education networks (NRENs)<sup>viii</sup> active in the country? (ref.: (iii) 18.c)

**YES** / NO

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

Switzerland's NREN, SWITCH, provides high-speed, secure networking for academic and research institutions. It connects major organizations such as CERN, SIB, SNSF, and Swiss universities while integrating Switzerland into international networks like GÉANT and e-IRG, facilitating seamless cross-border collaboration:

<https://www.switch.ch/en>

- 3.4 Are there national or institutional [open science infrastructures](#)<sup>ix</sup> in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

**YES / UNDER DEVELOPMENT / NO**

If yes or under development, please indicate which type(s) of infrastructure:

Most Swiss institutions have their own repositories, listing many of their publications. However, these repositories differ in their coverage of institutional research outputs and the completeness of metadata for available publications.

Switzerland's complex research system also operates a myriad of IT infrastructures, managed infrastructures, protocols, and standards. Because of their sheer quantity and diversity, no comprehensive overview of these infrastructures exists despite several attempts. The list below should be considered examples.

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.
  - CERN supports federated IT infrastructure for open science both nationally and internationally through platforms such as Zenodo (<https://zenodo.org/>), a free, non-commercial repository for data, software, and research artifacts, co-funded by the European Commission; Invenio (<https://invenio-software.org/>), an open-source library management system used by digital libraries worldwide; and Indico (<https://getindico.io/>): an open-source tool for conference and event management, used by 200+ institutions globally.
  - Worldwide LHC Computing Grid (WLCG): coordinated by CERN, it connects 170+ centers in 42 countries to support the massive data processing, computing, and storage needs of the Large Hadron Collider (LHC) experiments across countries:  
<https://home.cern/science/computing/grid>
  - SWITCH (see also 3.3) provides several key infrastructural services such as SWITCH Engines (<https://www.switch.ch/de/switch-engines>), a Swiss academic cloud service offering scalable computing and storage for research institutions; SWITCH Drive (<https://www.switch.ch/en/drive>), a secure cloud storage solution for researchers to store, share, and collaborate on data, ensuring compliance with Swiss data protection laws; and SWITCH edu-ID (<https://eduid.ch/>), a federated digital identity for Swiss universities and national service users, enabling secure, single-login access to data repositories, OA journals, and cloud tools.
  - Swiss National Supercomputing Centre (CSCS): develops and operates advanced high-performance computing systems, providing essential services for Swiss researchers in simulations and complex data analysis:  
<https://www.cscs.ch/>
  - Swiss National Data and Service Center for the Humanities (DaSCH): operates since 2021 as a national open data infrastructure primarily funded by the Swiss National Science Foundation (SNSF). Data stored with DaSCH is accessible by the FAIR-Principle:  
<https://www.dasch.swiss>
  - Swiss Centre of Expertise in the Social Sciences (FORS): provides social science researchers in Switzerland (and abroad) with various data services

(e.g., collection and analysis, archiving, management) and tools for information infrastructure:

<https://forscenter.ch>

All of the listed infrastructures are non-commercial as they are developed and maintained by academic, governmental, or publicly funded institutions with the goal of supporting open science, collaboration, and research infrastructure rather than commercial profit-making.

- community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society

- Materials Cloud: a service operated by the MARVEL National Centre of Competence whose Leading House is the Swiss Federal Institute of Technology Lausanne (EPFL), and is hosted at the Swiss National Supercomputing Center (CSCS) in Lugano, while being supported by an international community of researchers. Materials Cloud provides open-access computational tools, curated datasets, and educational resources for materials science, fostering collaboration and reproducibility in research, compliant with the FAIR principle:

<https://www.materialscloud.org/home>

- CERN Open Data Portal: managed by CERN, it fosters community engagement by providing open access to research data, software, and analysis tools from LHC experiments. The portal encourages contributions from researchers, educators, and the broader scientific community to promote transparency, reproducibility, and collaboration in high-energy physics:

<https://opendata.cern.ch/>

- Swiss Digital Research Infrastructure for Arts and Humanities (DARIAH-CH): aims to build a national network that enhances and supports digitally enabled research and teaching across the arts and humanities. The network fosters collaboration among Swiss researchers and institutions, as well as international partners. Coordinated by the DaSCH, member institutions include the universities of Basel, Bern, Lausanne, Geneva, and Zurich, the ETH Domain (EPFL, ETHZ Library), universities of applied sciences, and the Swiss National Museum:

<https://www.dariah.ch>

- CLARIN-CH: As part of the European CLARIN network, CLARIN-CH brings together Swiss researchers working on language-related issues and provides them with digital infrastructure for linguistic resources, including tools, web services, and knowledge centers, along with opportunities for collaboration in joint national and European projects. Aligned with the FAIR principles, it facilitates interoperability and data sharing across institutions and disciplines and supports multilingual and culturally diverse research in the humanities and social sciences:

<https://clarin-ch.ch>

- Linguistic Research Infrastructure (LiRI): based at the University of Zurich, LiRI provides advanced digital tools and services for linguistic research in

Switzerland. It supports researchers with data collection, processing, and analysis across various languages and disciplines:

<https://www.liri.uzh.ch/en.html>

- Swiss Institute of Bioinformatics (SIB): develops and maintains essential bioinformatics resources for life sciences research and data in Switzerland and beyond for academic and clinical research:

<https://www.sib.swiss>

- Swiss Personalized Health Network (SPHN): promotes the standardization and interoperability of health data across Swiss research institutions and hospitals:

<https://sphn.ch>

All of the listed infrastructures are **non-commercial**.

- platforms for exchanges and co-creation of knowledge between scientists and society

Citizen Science Zurich: hosted by the University of Zurich (UZH), this Swiss platform fosters collaboration and dialogue between researchers and citizen scientists to conceptually shape research methodologies and questions across various socially relevant sectors such as health, education, energy, and environment. For example, as part of the study “FICUS” (Family Support Intervention in Intensive Care Units), funded by the Swiss National Science Foundation, workshops are organized involving patients, family members, and health researchers to co-design a “mixed methods” evaluation of patient and public involvement (PPI) and partnership in health research. These exchanges help to define indicators and practical recommendations to improve future user-researcher collaboration and PPI, ultimately enhancing the relevance and effectiveness of health research for patients and their families.

Citizen Science Zurich is a **non-commercial** platform supported by public funding from UZH and ETHZ, and currently hosts over 30 active projects, some with open science resources like MOOCs and teaching materials:

<https://www.citizenscience.uzh.ch/en.html>

<https://www.citizenscience.uzh.ch/en/projects/ficus.html>

- community-based monitoring and information systems to complement national, regional and global data and information systems.

Citizen Science Zurich encourages community-level involvement in scientific research through active data collection and interpretation in projects across sectors. For example, the project “Swiss Photovoltaic Journeys” invites participation from homeowners with solar energy installations, to share data on specific questions (e.g., what initiated their interest in solar energy?; costs, problems, and electricity bills over the years). By collecting and sharing real-world experiences from photovoltaic system owners, the project aims to provide transparent data and insights to encourage wider adoption and optimize system performance:

<https://www.citizenscience.uzh.ch/en.html>

<https://www.citizenscience.uzh.ch/en/projects/solarstories.html>



**For each infrastructure**, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

**YES** / NO

**If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Yes, many infrastructures mentioned under 3.4 operate as federated regional or international OS infrastructures. It is particularly common for Swiss infrastructures to be nodes in European Union federated infrastructures such as the European Research Infrastructure Consortium (ERIC).

Switzerland participates in numerous ERICs, which allows Swiss researchers to make use of internationally coordinated research infrastructures while also coordinating the further development of their own scientific fields. Swiss membership of any one ERIC is ultimately decided by the Federal Council and voted by the ERIC General Assembly, while the SNSF and SERI are involved in membership procedures:

<https://www.sbf.admin.ch/sbf/en/home/research-and-innovation/international-cooperation-r-and-i/eric.html#-745063428>

Switzerland is currently active in six ERICs, including the DARIAH ERIC (Swiss node: Swiss National Data and Service Center for the Humanities DaSCH). Switzerland also holds "observer" status in four ERICs, including the European Social Survey (ESSurvey) ERIC (Swiss node: Swiss Centre of Expertise in the Social Sciences FORS) and the CLARIN ERIC (Swiss node: LiRI @ University of Zurich / CLARIN-CH).

Swiss infrastructures are also commonly embedded in other, informal networks of infrastructures at regional or international level, that operate with a level of federation.

**If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

YES / NO

- 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

**YES** / NO

**If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas,

elements of open science<sup>1</sup>, stage of research, accessibility to all the relevant users in the region or internationally).

Through institutions like the Swiss National Supercomputing Centre (CSCS), Swissuniversities Development and Cooperation Network (SUDAC), and initiatives such as Excellence in Africa (EXAF), Switzerland supports multinational research efforts, open-access data, and digital education. These partnerships contribute to global scientific progress, including in the Global South, by providing advanced computational resources, fostering innovation, and aligning with the U.N. Sustainable Development Goals (SDGs). Key initiatives include:

- CSCS: as part of ETH Zurich, it provides data infrastructure for the Square Kilometre Array Observatory (SKAO)—the world's largest radio telescope, with arrays in South Africa and Australia, and the Cherenkov Telescope Array Observatory (CTAO)—a next-generation gamma-ray observatory with a key site in Chile:  
<https://www.cscs.ch/>
- SUDAC: seeks targeted support for collaboration between the various types of Swiss higher education institutions and their partners from the Global South to achieve an excellent standard of education, research and innovation on global challenges. To achieve its aims, the SUDAC programme has promoted the development of two instruments:
  - 'Consortia for Education and Research (COFER)' that are oriented towards the Sustainable Development Goals (SDGs) of the U.N. Agenda 2030. COFER MOOLS, a joint project by EPFL and HES-SO, enhances digital education through online labs in countries like Iran, Niger, Lebanon, and Djibouti.
  - Clusters of Cooperation in the Global South' (CLOCs) are expected to serve as 'regional hubs', connecting a variety of 'nodes', such as existing research and education partnerships, regionally active COFER, local knowledge platforms and institutions.  
<https://www.swissuniversities.ch/it/themen/entwicklung-und-zusammenarbeit/p-6-swissuniversities-development-and-cooperation-network-sudac>
- EXAF: Launched in 2019 by EPFL, EXAF advances digital research and education, offering African STEM students high-quality resources and remote experimentation via Open Online Labs.  
<https://www.epfl.ch/research/domains/exaf/>

#### 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

- 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

**YES** / NO

**If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- Structured capacity-building initiatives on open science within higher education institutions are supported by the swissuniversities PgB Open Science II instrument, which, for example, provides funding for training ORD specialists and data stewards across Swiss universities:  
<https://www.swissuniversities.ch/themen/open-science/programm-open-science>
- The ETH Domain ORD program, which focuses on advancing ORD practices within the ETH Domain institutions, aims to enhance ORD knowledge, management, infrastructure, and skills through five defined measures:  
<https://ethrat.ch/en/eth-domain/open-research-data/>
- National initiatives have supported relevant networks to foster informal capacity building, particularly for data stewards and research software engineers. The longstanding Swiss Library Network for Education and Research has been instrumental in building Open Science capacity within libraries:  
<https://www.swissuniversities.ch/en/organisation/bodies/networks-and-agencies/swiss-library-network-for-education-and-research>

- 4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

YES / **NO**

**If yes**, please provide more information and links as relevant.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Targeted capacity-building programmes on for STI policymakers remains an area that requires further development. Obstacles may include competing priorities within policymaking bodies as well as a gap in awareness in these communities regarding the importance of including Open Science into national policies.

- 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

YES / **NO**

**If yes**, please provide more information and links as relevant.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No national framework exists. However, individual higher education institutions have integrated aspects of open science competencies (e.g., research data management, OA publishing) into their curricula, primarily at the doctoral level.

- 4.4 Are open educational resources<sup>x</sup> as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

**YES** / PARTLY / NO

Several OER initiatives exist in Switzerland.

- The SwissMOOCService, founded by ETH Zurich and EPFL and hosted on Switch infrastructure, provides access to a wide range of online courses across disciplines from leading institutions, available in multiple languages. Additionally, it offers advanced course development and management tools for instructors and course developers:  
<https://www.swissmooc.ch>
- Switch OER, Switzerland's OER repository for higher education institutions, also offers diverse educational materials across disciplines, from natural sciences to arts and humanities:  
<https://oer.switch.ch/>
- On both of these platforms, courses and OERs relevant to OS practices can be found. For example, Switch OER offers the training course "Didactic support to introduce the issue of Creative Commons in the context of sharing educational resources":  
<https://oer-repository.switch.ch/edu-sharing/components/render/6ef6da1d-842e-499f-8f8f-61d586c1a652?q=open%20science>
- Additionally, the ETH Domain ORD program uses Zenodo as the OER platform in an initiative to collect all ORD-related training materials from participating institutions in the form of OERs. These materials will then be used for an open MOOC to be released on SwissMOOC (ongoing).

- 4.5 Have there been any initiatives in your country to support science communication in line with open science values and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

**YES** / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Switzerland actively promotes science communication to foster scientific dialogue with multiple audiences. Examples include:

- The Swiss National Science Foundation (SNSF) supports science communication between researchers as well as communication between researchers and the public. It therefore awards grants for scientific meetings and publications as well as for projects that foster dialogue between science and society:  
<https://www.snf.ch/en/EXrMdruJNhNtvdMg/page/funding/science-communication>
- Following the heightened demand for public engagement with science after COVID-19, the Swiss Academies of Arts and Sciences formed the expert group "Communicating Sciences and Arts in Times of Digital Media". Representing diverse disciplines, institutions, and linguistic regions, the group assessed science communication in Switzerland. In 2021, they published a

comprehensive report highlighting its strengths and providing 20 recommendations to enhance public engagement and embed science communication into scientific culture and practice:

<https://akademien-schweiz.ch/en/themen/science-communication/>

- Pint of Science: an annual event that brings researchers to local bars and cafés to share cutting-edge discoveries in their field with the public. The next edition is scheduled in May 2025:  
<https://pintofscience.ch/>
- In addition to this physical event, Pint of Science is also accessible as a podcast (<https://pintofscience.ch/podcast>) and short videos (*Shot of Science*, <https://pintofscience.ch/shots-of-science>) .
- Swiss higher education institutions offer various courses to help young researchers communicate science effectively.
- Communication initiatives are also in place for specific scientific fields, such as [ZettaBytes](#) (EPFL), which promotes and explains big ideas of computer science to a general audience.

## 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\) 20.b, 20c, 20i](#))

**YES** / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- Currently, 19 Swiss institutions have signed the Coalition for Advancing Research Assessment (CoARA), including: ETH Domain research institutions such as EPFL and ETH Zurich, major universities (e.g., University of Geneva, University of Fribourg, and University of Lausanne), universities of applied sciences in Bern and Zurich, Swiss academies of arts and sciences, universities of teacher education, the university umbrella organization, swissuniversities, and Switzerland's primary scientific research funder, the Swiss National Science Foundation (SNSF). As part of their commitment to this international initiative, and in alignment with the Agreement on Reforming Research Assessment, each Swiss signatory is expected to submit an institutional Action Plan. These plans will outline how they intend to align their research evaluation practices to CoARA's principles, of which Open Science, and in particular the recognition of the diversity of contributions, is an important component, while integrating them within existing institutional frameworks:  
<https://coara.eu/agreement/the-agreement-full-text/>  
<https://coara.eu/agreement/signatories/?category%5B0%5D=switzerland#signatories>
- In addition to institutional-level engagement, the Swiss National Chapter of CoARA unites Swiss universities and funding agencies, encompassing Switzerland's diverse language regions and higher education landscape. It is expected to foster a more coordinated and systematic approach to research evaluation reform, aligning with European and international developments:

<https://coara.eu/working-groups/national-chapters/coara-national-chapter-switzerland/>

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

**YES** / NO

If yes, please provide details and links as relevant.

From 2022, the Swiss National Science Foundation (SNSF) introduced a “narrative CV” across its funding schemes. This CV format gives more flexibility to researchers to explain their scientific achievements beyond mere publications and explicitly includes “contributions to (i) the generation of knowledge, innovation, and advancement of research, (ii) the development and support of peers and the wider research community, and (iii) Broader Society”.

<https://www.snf.ch/en/gKcnwW6aEft4bMPF/page/your-curriculum-vitae-all-about-the-cv-format>

- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

**YES** / NO

If yes, for each initiative, please provide more information, including on the stakeholders<sup>xi</sup> that are involved or lead the initiative, and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- Since 2023, the Academies of Sciences Switzerland have been awarding the National Prize for Open Research Data (ORD) as part of the ORD Action Plan, implemented under the National Strategy for Open Research Data. This prize recognizes researchers for innovative practices in open research and encourages the transition to more transparent and collaborative research methods. For the 2024 competition, the prize focused on two key areas: (i) collaborative and/or interdisciplinary reuse of research data, and (ii) educational and/or outreach activities related to data reuse. The four winning projects were: Pathoplexus (University of Basel), ROADS (Zurich University of Applied Sciences, ZHAW), openwashdata (a global initiative led by ETH Zurich), and ENAC-IT4Research (EPFL). The ORD Prize is expected to be re-awarded in 2025.

<https://ord.swiss-academies.ch/national-ord-prize/overview>

- The SNSF actively supports open access (OA) and open research data (ORD) through funding requirements. While the SNSF does not yet have a dedicated open science award, it provides funding incentives for researchers implementing open science principles.

<https://www.snf.ch/en/VyUvGzptStOEpuoC/topic/open-access-publications>

<https://www.snf.ch/en/dMILj9t4LNk8NwyR/topic/open-research-data>

- 5.4 Have there been any unintended negative consequences<sup>xii</sup> of open science practices reported in your country? (ref.: (v) 20)

YES / **NO**

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

## 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\) 21](#))

YES, at the national level.

YES, at the institutional level.

**YES**, at both national and institutional levels.

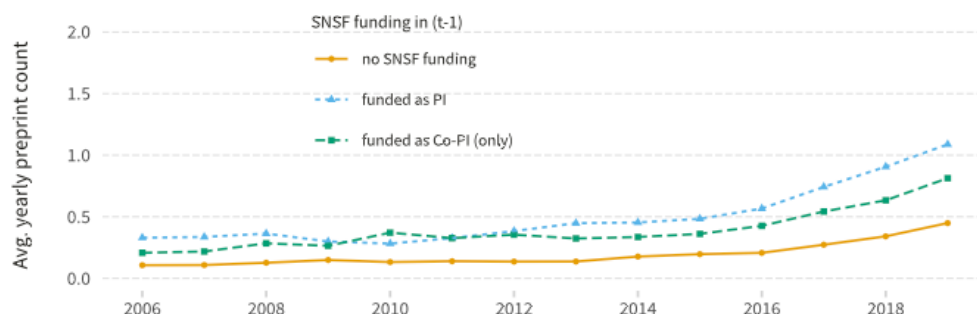
NO

If **yes**, please indicate which of the recommended actions they address and provide more information and links as relevant:

- Promotion of pre-prints: In accordance with Switzerland's National Open Access Strategy, the Swiss National Science Foundation (SNSF) mandates open access for all publications resulting from publicly funded research, supporting dissemination through various open-access channels, including preprints. While no explicit policies on preprints exist, SNSF-funded researchers are clearly engaged in the preprint culture:

[https://www.nature.com/articles/s41599-021-00891-x?utm\\_source=chatgpt.com](https://www.nature.com/articles/s41599-021-00891-x?utm_source=chatgpt.com)

<https://www.snf.ch/en/MDecEyLJgpSTk0cU/page/open-access-information-for-researchers>



- Valuing and sharing of negative scientific results and associated data: the Swiss National Open Research Data (ORD) Strategy and Action Plan aim to create optimal conditions for the access, management, and reuse of publicly financed research data aligned with the FAIR principles, which naturally includes the sharing of negative scientific results and data.

<https://www.swissuniversities.ch/en/topics/open-science/open-research-data/national-strategy>

- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects: development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas: While specific initiatives targeting marginalized communities are not explicitly detailed in the available sources, platforms like Citizen Science Zurich, based at the University of Zurich (UZH) in collaboration with ETH Zurich, promote participatory approaches that inherently encourage inclusivity. These methods help integrate pressing social issues into research across various sectors, including health,



energy, environment, and education, by involving a diverse range of participants. This includes cancer survivors, youth and elderly individuals with specific conditions (e.g., hearing loss, mental illness), and women, ensuring that research reflects and addresses their unique experiences and challenges. Moreover, indigenous communities play a vital role in participatory research, such as in anthropology. For instance, a project at the Ethnographic Museum at the University of Zurich engaged Wounaan people from Colombia and Panama to explore artifacts from their culture. Their contributions were instrumental in deepening the understanding of material culture, traditional knowledge, and the role of artisan collections, fostering meaningful dialogue between researchers and originator societies.

<https://www.citizenscience.uzh.ch/en/projects.html>

<https://www.citizenscience.uzh.ch/en/projects/wounaan.html>

- promoting open innovation practices.
  - The Open Science Programme II of swissuniversities, scheduled to run from 2025 to 2026, aims to strategically advance open innovation across three key dimensions, supported by project-related federal contributions (PgB) totalling CHF 6.45 million. It will do so by:  
Enhancing capacities for scientific collaboration through the implementation of the goals and measures of:  
(i) the Swiss National OA Strategy; and  
(ii) the Swiss National ORD Strategy; and by  
(iii) promoting innovative areas for Open Science, specifically designed to facilitate knowledge and technology transfer between universities, businesses, and society, as well as by fostering science diplomacy and international collaboration, thereby broadening the impact of open innovation both nationally and globally:  
[https://www.swissuniversities.ch/fileadmin/swissuniversities/Dokumente/Hochschulpolitik/Open\\_Science/2025/PgB-Open-Science-II-2025-2028-Implementationsplan.pdf](https://www.swissuniversities.ch/fileadmin/swissuniversities/Dokumente/Hochschulpolitik/Open_Science/2025/PgB-Open-Science-II-2025-2028-Implementationsplan.pdf)
  - Innosuisse, the Swiss Innovation Agency, supports open innovation by fostering collaboration between researchers, start-ups, and established companies. For example, the first generation of thematic 'Innovation Booster' initiatives supported by Innosuisse was concluded at the end of 2024. Launched in 2021, these thematic initiatives which serve as a springboard for promoting a culture of co-creation and collaboration across innovation hubs like Energy Lab, Swiss Food Ecosystems, and Blockchain Nation Switzerland, have made a significant contribution to redefining the approach to innovation in Switzerland. The first signs of the long-term impact these initiatives is seen in local projects such as E4citizens, aiming to initiate discussions and collaboration around greener energy practices within local organizations and communities using a gamified tool, testifying to the democratization of open innovation in a number of sectors. In addition, programmes such as the diversity-focused Women's in Energy Lab & Life (WELL) are demonstrating their ability to address societal challenges, such as the underrepresentation of women in technical fields:  
<https://www.innosuisse.admin.ch/en/end-of-the-first-innovation-booster>  
<https://energylivinglab.com/departments/e4citizens/>  
[https://energylab.site/wp-content/uploads/2021/01/Gender\\_equity\\_EnergyLab1-1.pdf](https://energylab.site/wp-content/uploads/2021/01/Gender_equity_EnergyLab1-1.pdf)

- Innosuisse also runs innovation cheques worth up to CHF 15k to help Small and medium-sized enterprises (SMEs) connect with research institutions for open innovation by assessing the viability, market potential, and collaboration potential of their innovation concepts:  
<https://www.innosuisse.admin.ch/en/innovation-cheque>

- Others (with a box to add text)

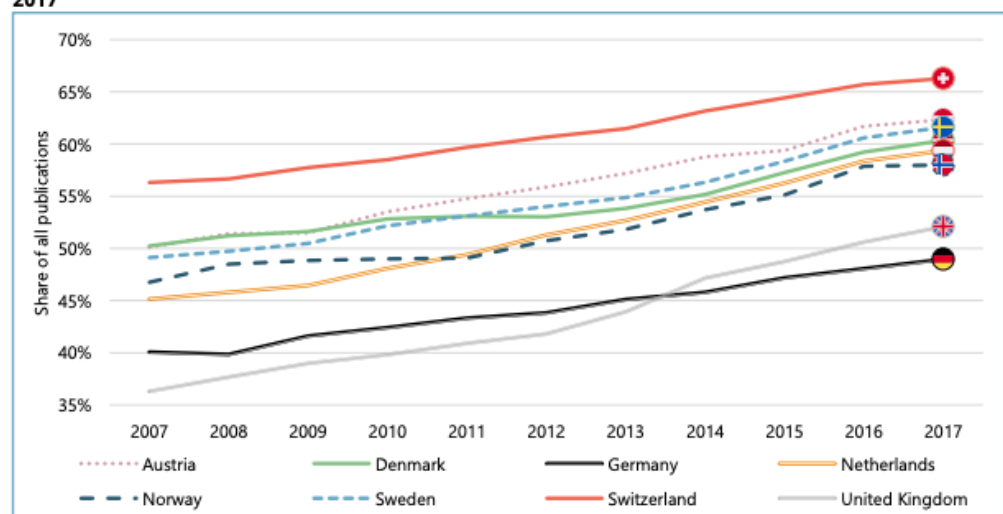
If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

## 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

### 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Swiss research and its higher education landscape are highly connected at the international level. It has become even more so in recent years with a constant increase in student mobility, internationalization of university teaching staff, and the interest of researchers in cooperating with European and global partners. For example, an evaluation prepared for the State Secretariat for Education, Research and Innovation (SERI) in 2020 on Switzerland's bilateral cooperation programmes in science and technology showed that, during the period 2007-2017, among eight European countries (Austria, Denmark, Germany, the Netherlands, Norway, Sweden, Switzerland and the United Kingdom), Switzerland had the highest share of collaborative publications with all countries every year. In 2017, this share was 66%—that is, two-thirds of all Swiss publications were prepared in collaboration with researchers from at least one foreign country.

**Figure 7.3. Growth in collaborative publications with all countries as a share of all publications, 2007-2017**



Source: Analysis based on the Scopus database

Link: <https://irisgroup.dk/wp-content/uploads/2020/03/EVALUA1.pdf>

This openness in collaboration naturally extends to and demonstrates Switzerland's strong engagement in OS practices, OS initiatives and

infrastructures, etc. The necessary support for international collaborations is provided by the SERI:

<https://www.sbf.admin.ch/sbf/en/home/research-and-innovation/international-cooperation-r-and-i.html>

- 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

YES / **NO**

If **yes**, please provide links as relevant and more information.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific strategy exists, however Swiss actors are involved in many European and international networks, as well as the EU-wide framework programme Horizon Europe (see 7.3), which facilitate the natural exchange of best practices, capacity building, and metrics for open science. Even without a formal strategy or plan, these activities are still taking place.

- 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

YES / **NO**

If **yes**, please provide links as relevant and more information.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Switzerland is not involved, to the best of our knowledge, in such discussions. However, Switzerland participates in the EU Framework Programme, the most recent edition of which, Horizon Europe (2021–2027, EUR 95.5 billion), is the largest research and innovation funding programme in the world. This programme contains several mechanisms for open science funding, in which Switzerland is therefore a party. For example, Horizon Europe mandates open access to publications and research data and recommends other open practices as appropriate, such as open peer review and the inclusion of citizens as key knowledge actors.

Links:

<https://www.sbf.admin.ch/sbf/en/home/research-and-innovation/international-cooperation-r-and-i/eu-framework-programmes-for-research/horizon-europe.html#-1558872795>

[https://rea.ec.europa.eu/open-science\\_en](https://rea.ec.europa.eu/open-science_en)

[https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/programme-guide\\_horizon\\_en.pdf](https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/programme-guide_horizon_en.pdf) (page 40)

## 8. GENERAL CONSIDERATIONS

- 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

**YES**/NO

If **yes**, please provide more information.

Switzerland is a federal country with a decentralized culture of governance. This extends to the research sector. The UNESCO Recommendation cannot be implemented as a single, top-down process, but requires that all actors in the Swiss research ecosystem (ministry, national-level entities, higher-education institutions), engage with it and implement it at their level. The answers in this questionnaire reflect this reality.

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- i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).
  - ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.  
Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.  
Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.
  - iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.
  - iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
  - v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
  - vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
  - vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
  - viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.
  - ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.
  - x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))
  - xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.
  - xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

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